

1. Administrative & Study Identification

Full Study Title: A Phase 3, randomized, parallel-group, placebo-controlled, double-blinded (third-party unblinded) study to evaluate the safety, tolerability, and immunogenicity of different vaccination schedules of the 6-valent OspA-based Lyme disease (LD) vaccine, VLA15, in healthy participants 18 through 44 years of age. **Short Title/Acronym:** A Study to Learn About Different Dosing Schedules of a Lyme Disease Vaccine in Healthy Adults **Protocol Number:** C4601019 **NCT Number:** NCT07226882 **Sponsor Name:** Pfizer **Investigational Product:** VLA15 (6-valent OspA-based Lyme disease vaccine) and Placebo (saline) **Phase of Development:** Phase III **Medical Monitor:** Pfizer Clinical Operations / External Data Monitoring Committee (EDMC)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the immunogenicity (Geometric Mean Concentration [GMC] and Geometric Mean Fold Rise [GMFR] of Anti-OspA IgG) and safety of different vaccination schedules of the VLA15 Lyme disease vaccine. **Secondary Objectives:** To further assess the overall safety, tolerability, and additional immune responses (seroconversion rates) across the parallel dosing groups over the 2-year duration.

2.2 Background & Rationale To address the growing public health concern of Lyme borreliosis, the most common tick-borne disease in the northern hemisphere. Currently, no human vaccines are available for its prevention. This study aims to learn about the safety and immune effects of the VLA15 vaccine when administered at different time points, ultimately identifying the optimal dosing schedule to maximize immunogenicity while maintaining an acceptable safety profile in healthy adults.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Clinical Trial) **Control Type:** Placebo-controlled (Saline) **Blinding:** Double-blinded (third-party unblinded) **Randomization:** 1:1:1:1 ratio into 4 parallel study groups

3.2 Planned Enrollment & Duration Target **SN\$:** 200 participants **Number of Sites:** Multiple study clinics (United States) **Estimated Study Start:** 19-Nov-2025 **Estimated Primary Completion:** 14-Feb-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy individuals between 18 and 44 years of age, as determined by medical history and clinical judgment. 2. Have never received a vaccine for Lyme disease prior to the study. 3. Willing and able to comply with all scheduled visits, investigational product receipt, and other procedures throughout the approximately 2-year study duration.

4.2 Exclusion Criteria 1. Pregnant or breastfeeding participants, or fertile individuals unwilling to use effective contraceptive methods. 2. Allergies or contraindications to vaccines or their components. 3. Health issues including blood clotting deficiencies, immunodeficiencies, bone marrow disorders, or uncontrolled psychiatric conditions. 4. Recent or concurrent receipt of therapies to treat malignancies, blood/plasma products, immunoglobulins, systemic corticosteroids, immunosuppressants, or anticoagulant therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Total of 5 doses per participant. Four (4) doses will be the active study vaccine (VLA15) and one (1) dose will be a placebo (saline). The specific visit at which the placebo is administered (Visit 3, 5, 7, or 9) varies based on the randomized group schedule. **Route of Administration:** Intramuscular (IM) injection in the upper arm. **Duration of Treatment:** Approximately 2 years (encompassing the primary series and evaluations).

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications not affecting immune response or blood coagulation. **Prohibited:** Systemic corticosteroids, immunosuppressants, therapies for malignancies, blood/plasma products, immunoglobulins, and anticoagulant therapy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Day -28 to 0	Informed Consent, Medical History, Health Check, Blood Draw
2	Day 0 to 28	Vaccination
3	Day 28 to 56	Health Check, Blood Draw
4	Day 56 to 84	Health Check, Blood Draw
5	Day 84 to 112	Health Check, Blood Draw
6	Day 112 to 140	Health Check, Blood Draw
7	Day 140 to 168	Health Check, Blood Draw
8	Day 168 to 196	Health Check, Blood Draw
9	Day 196 to 224	Health Check, Blood Draw
10	Day 224 to 252	Health Check, Blood Draw
11	Day 252 to 280	Health Check, Blood Draw
12	Day 280 to 308	Health Check, Blood Draw
13	Day 308 to 336	Health Check, Blood Draw
14	Day 336 to 364	Health Check, Blood Draw
15	Day 364 to 392	Health Check, Blood Draw
16	Day 392 to 420	Health Check, Blood Draw
17	Day 420 to 448	Health Check, Blood Draw
18	Day 448 to 476	Health Check, Blood Draw
19	Day 476 to 504	Health Check, Blood Draw
20	Day 504 to 532	Health Check, Blood Draw
21	Day 532 to 560	Health Check, Blood Draw
22	Day 560 to 588	Health Check, Blood Draw
23	Day 588 to 616	Health Check, Blood Draw
24	Day 616 to 644	Health Check, Blood Draw
25	Day 644 to 672	Health Check, Blood Draw
26	Day 672 to 700	Health Check, Blood Draw
27	Day 700 to 728	Health Check, Blood Draw
28	Day 728 to 756	Health Check, Blood Draw
29	Day 756 to 784	Health Check, Blood Draw
30	Day 784 to 812	Health Check, Blood Draw
31	Day 812 to 840	Health Check, Blood Draw
32	Day 840 to 868	Health Check, Blood Draw
33	Day 868 to 896	Health Check, Blood Draw
34	Day 896 to 924	Health Check, Blood Draw
35	Day 924 to 952	Health Check, Blood Draw
36	Day 952 to 980	Health Check, Blood Draw
37	Day 980 to 1008	Health Check, Blood Draw
38	Day 1008 to 1036	Health Check, Blood Draw
39	Day 1036 to 1064	Health Check, Blood Draw
40	Day 1064 to 1092	Health Check, Blood Draw
41	Day 1092 to 1120	Health Check, Blood Draw
42	Day 1120 to 1148	Health Check, Blood Draw
43	Day 1148 to 1176	Health Check, Blood Draw
44	Day 1176 to 1204	Health Check, Blood Draw
45	Day 1204 to 1232	Health Check, Blood Draw
46	Day 1232 to 1260	Health Check, Blood Draw
47	Day 1260 to 1288	Health Check, Blood Draw
48	Day 1288 to 1316	Health Check, Blood Draw
49	Day 1316 to 1344	Health Check, Blood Draw
50	Day 1344 to 1372	Health Check, Blood Draw
51	Day 1372 to 1400	Health Check, Blood Draw
52	Day 1400 to 1428	Health Check, Blood Draw
53	Day 1428 to 1456	Health Check, Blood Draw
54	Day 1456 to 1484	Health Check, Blood Draw
55	Day 1484 to 1512	Health Check, Blood Draw
56	Day 1512 to 1540	Health Check, Blood Draw
57	Day 1540 to 1568	Health Check, Blood Draw
58	Day 1568 to 1596	Health Check, Blood Draw
59	Day 1596 to 1624	Health Check, Blood Draw
60	Day 1624 to 1652	Health Check, Blood Draw
61	Day 1652 to 1680	Health Check, Blood Draw
62	Day 1680 to 1708	Health Check, Blood Draw
63	Day 1708 to 1736	Health Check, Blood Draw
64	Day 1736 to 1764	Health Check, Blood Draw
65	Day 1764 to 1792	Health Check, Blood Draw
66	Day 1792 to 1820	Health Check, Blood Draw
67	Day 1820 to 1848	Health Check, Blood Draw
68	Day 1848 to 1876	Health Check, Blood Draw
69	Day 1876 to 1904	Health Check, Blood Draw
70	Day 1904 to 1932	Health Check, Blood Draw
71	Day 1932 to 1960	Health Check, Blood Draw
72	Day 1960 to 1988	Health Check, Blood Draw
73	Day 1988 to 2016	Health Check, Blood Draw
74	Day 2016 to 2044	Health Check, Blood Draw
75	Day 2044 to 2072	Health Check, Blood Draw
76	Day 2072 to 2100	Health Check, Blood Draw
77	Day 2100 to 2128	Health Check, Blood Draw
78	Day 2128 to 2156	Health Check, Blood Draw
79	Day 2156 to 2184	Health Check, Blood Draw
80	Day 2184 to 2212	Health Check, Blood Draw
81	Day 2212 to 2240	Health Check, Blood Draw
82	Day 2240 to 2268	Health Check, Blood Draw
83	Day 2268 to 2296	Health Check, Blood Draw
84	Day 2296 to 2324	Health Check, Blood Draw
85	Day 2324 to 2352	Health Check, Blood Draw
86	Day 2352 to 2380	Health Check, Blood Draw
87	Day 2380 to 2408	Health Check, Blood Draw
88	Day 2408 to 2436	Health Check, Blood Draw
89	Day 2436 to 2464	Health Check, Blood Draw
90	Day 2464 to 2492	Health Check, Blood Draw
91	Day 2492 to 2520	Health Check, Blood Draw
92	Day 2520 to 2548	Health Check, Blood Draw
93	Day 2548 to 2576	Health Check, Blood Draw
94	Day 2576 to 2604	Health Check, Blood Draw
95	Day 2604 to 2632	Health Check, Blood Draw
96	Day 2632 to 2660	Health Check, Blood Draw
97	Day 2660 to 2688	Health Check, Blood Draw
98	Day 2688 to 2716	Health Check, Blood Draw
99	Day 2716 to 2744	Health Check, Blood Draw
100	Day 2744 to 2772	Health Check, Blood Draw
101	Day 2772 to 2800	Health Check, Blood Draw
102	Day 2800 to 2828	Health Check, Blood Draw
103	Day 2828 to 2856	Health Check, Blood Draw
104	Day 2856 to 2884	Health Check, Blood Draw
105	Day 2884 to 2912	Health Check, Blood Draw
106	Day 2912 to 2940	Health Check, Blood Draw
107	Day 2940 to 2968	Health Check, Blood Draw
108	Day 2968 to 2996	Health Check, Blood Draw
109	Day 2996 to 3024	Health Check, Blood Draw
110	Day 3024 to 3052	Health Check, Blood Draw
111	Day 3052 to 3080	Health Check, Blood Draw
112	Day 3080 to 3108	Health Check, Blood Draw
113	Day 3108 to 3136	Health Check, Blood Draw
114	Day 3136 to 3164	Health Check, Blood Draw
115	Day 3164 to 3192	Health Check, Blood Draw
116	Day 3192 to 3220	Health Check, Blood Draw
117	Day 3220 to 3248	Health Check, Blood Draw
118	Day 3248 to 3276	Health Check, Blood Draw
119	Day 3276 to 3304	Health Check, Blood Draw
120	Day 3304 to 3332	Health Check, Blood Draw
121	Day 3332 to 3360	Health Check, Blood Draw
122	Day 3360 to 3388	Health Check, Blood Draw
123	Day 3388 to 3416	Health Check, Blood Draw
124	Day 3416 to 3444	Health Check, Blood Draw
125	Day 3444 to 3472	Health Check, Blood Draw
126	Day 3472 to 3500	Health Check, Blood Draw
127	Day 3500 to 3528	Health Check, Blood Draw
128	Day 3528 to 3556	Health Check, Blood Draw
129	Day 3556 to 3584	Health Check, Blood Draw
130	Day 3584 to 3612	Health Check, Blood Draw
131	Day 3612 to 3640	Health Check, Blood Draw
132	Day 3640 to 3668	Health Check, Blood Draw
133	Day 3668 to 3696	Health Check, Blood Draw
134	Day 3696 to 3724	Health Check, Blood Draw
135	Day 3724 to 3752	Health Check, Blood Draw
136	Day 3752 to 3780	Health Check, Blood Draw
137	Day 3780 to 3808	Health Check, Blood Draw
138	Day 3808 to 3836	Health Check, Blood Draw
139	Day 3836 to 3864	Health Check, Blood Draw
140	Day 3864 to 3892	Health Check, Blood Draw
141	Day 3892 to 3920	Health Check, Blood Draw
142	Day 3920 to 3948	Health Check, Blood Draw
143	Day 3948 to 3976	Health Check, Blood Draw
144	Day 3976 to 4004	Health Check, Blood Draw
145	Day 4004 to 4032	Health Check, Blood Draw
146	Day 4032 to 4060	Health Check, Blood Draw
147	Day 4060 to 4088	Health Check, Blood Draw
148	Day 4088 to 4116	Health Check, Blood Draw
149	Day 4116 to 4144	Health Check, Blood Draw
150	Day 4144 to 4172	Health Check, Blood Draw
151	Day 4172 to 4200	Health Check, Blood Draw
152	Day 4200 to 4228	Health Check, Blood Draw
153	Day 4228 to 4256	Health Check, Blood Draw
154	Day 4256 to 4284	Health Check, Blood Draw
155	Day 4284 to 4312	Health Check, Blood Draw
156	Day 4312 to 4340	Health Check, Blood Draw
157	Day 4340 to 4368	Health Check, Blood Draw
158	Day 4368 to 4396	Health Check, Blood Draw
159	Day 4396 to 4424	Health Check, Blood Draw
160	Day 4424 to 4452	Health Check, Blood Draw
161	Day 4452 to 4480	Health Check, Blood Draw
162	Day 4480 to 4508	Health Check, Blood Draw
163	Day 4508 to 4536	Health Check, Blood Draw
164	Day 4536 to 4564	Health Check, Blood Draw
165	Day 4564 to 4592	Health Check, Blood Draw
166	Day 4592 to 4620	Health Check, Blood Draw
167	Day 4620 to 4648	Health Check, Blood Draw
168	Day 4648 to 4676	Health Check, Blood Draw
169	Day 4676 to 4704	Health Check, Blood Draw
170	Day 4704 to 4732	Health Check, Blood Draw
171	Day 4732 to 4760	Health Check, Blood Draw
172	Day 4760 to 4788	Health Check, Blood Draw
173	Day 4788 to 4816	Health Check, Blood Draw
174	Day 4816 to 4844	Health Check, Blood Draw
175	Day 4844 to 4872	Health Check, Blood Draw
176	Day 4872 to 4900	Health Check, Blood Draw
177	Day 4900 to 4928	Health Check, Blood Draw
178	Day 4928 to 4956	Health Check, Blood Draw
179	Day 4956 to 4984	Health Check, Blood Draw
180	Day 4984 to 5012	Health Check, Blood Draw
181	Day 5012 to 5040	Health Check, Blood Draw
182	Day 5040 to 5068	Health Check, Blood Draw
183	Day 5068 to 5096	Health Check, Blood Draw
184	Day 5096 to 5124	Health Check, Blood Draw
185	Day 5124 to 5152	Health Check, Blood Draw
186	Day 5152 to 5180	Health Check, Blood Draw
187	Day 5180 to 5208	Health Check, Blood Draw
188	Day 5208 to 5236	Health Check, Blood Draw
189	Day 5236 to 5264	Health Check, Blood Draw
190	Day 5264 to 5292	Health Check, Blood Draw
191	Day 5292 to 5320	Health Check, Blood Draw
192	Day 5320 to 5348	Health Check, Blood Draw
193	Day 5348 to 5376	Health Check, Blood Draw
194	Day 5376 to 5404	Health Check, Blood Draw
195	Day 5404 to 5432	Health Check, Blood Draw
196	Day 5432 to 5460	Health Check, Blood Draw
197	Day 5460 to 5488	Health Check, Blood Draw
198	Day 5488 to 5516	Health Check, Blood Draw
199	Day 5516 to 5544	Health Check, Blood Draw
200	Day 5544 to 5572	Health Check, Blood Draw
201	Day 5572 to 5600	Health Check, Blood Draw
202	Day 5600 to 5628	Health Check, Blood Draw
203	Day 5628 to 5656	Health Check, Blood Draw
204	Day 5656 to 5684	Health Check, Blood Draw
205	Day 5684 to 5712	Health Check, Blood Draw
206	Day 5712 to 5740	Health Check, Blood Draw
207	Day 5740 to 5768	Health Check, Blood Draw
208	Day 5768 to 5796	Health Check, Blood Draw
209	Day 5796 to 5824	Health Check, Blood Draw
210	Day 5824 to 5852	Health Check, Blood Draw
211	Day 5852 to 5880	Health Check, Blood Draw
212	Day 5880 to 5908	Health Check, Blood Draw
213	Day 5908 to 5936	Health Check, Blood Draw
214	Day 5936 to 5964	Health Check, Blood Draw
215	Day 5964 to 5992	Health Check, Blood Draw
216	Day 5992 to 6020	Health Check, Blood Draw
217	Day 6020 to 6048	Health Check, Blood Draw
218	Day 6048 to 6076	Health Check, Blood Draw
219	Day 6076 to 6104	Health Check, Blood Draw
220	Day 6104 to 6132	Health Check, Blood Draw
221	Day 6132 to 6160	Health Check, Blood Draw
222	Day 6160 to 6188	Health Check, Blood Draw
223	Day 6188 to 6216	Health Check, Blood Draw
224	Day 6216 to 6244	Health Check, Blood Draw
225	Day 6244 to 6272	Health Check, Blood Draw
226	Day 6272 to 6300	Health Check, Blood Draw
227	Day 6300 to 6328	Health Check, Blood Draw
228	Day 6328 to 6356	Health Check, Blood Draw
229	Day 6356 to 6384	Health Check, Blood Draw
230	Day 6384 to 6412	Health Check, Blood Draw
231	Day 6412 to 6440	Health Check, Blood Draw
232	Day 6440 to 6468	Health Check, Blood Draw
233	Day 6468 to 6496	Health Check, Blood Draw
234	Day 6496 to 6524	Health Check, Blood Draw
235	Day 6524 to 6552	Health Check, Blood Draw
236	Day 6552 to 6580	Health Check, Blood Draw
237	Day 6580 to 6608	Health Check, Blood Draw
238	Day 6608 to 6636	Health Check, Blood Draw
239	Day 6636 to 6664	Health Check, Blood Draw
240	Day 6664 to 6692	Health Check, Blood Draw
241	Day 6692 to 6720	Health Check, Blood Draw
242	Day 6720 to 6748	Health Check, Blood Draw
243	Day 6748 to 6776	Health Check, Blood Draw
244	Day 6776 to 6804	Health Check, Blood Draw
245	Day 6804 to 6832	Health Check, Blood Draw
246	Day 6832 to 6860	Health Check, Blood Draw
247	Day 6860 to 6888	Health Check, Blood Draw
248	Day 6888 to 6916	Health Check, Blood Draw
249	Day 6916 to 6944	Health Check, Blood Draw
250	Day 6944 to 6972	Health Check, Blood Draw
251	Day 6972 to 7000	Health Check, Blood Draw
252	Day 7000 to 7028	Health Check, Blood Draw
253	Day 7028 to 7056	Health Check, Blood Draw
254	Day 7056 to 7084	Health Check, Blood Draw
255	Day 7084 to 7112	Health Check, Blood Draw
256	Day 7112 to 7140	Health Check, Blood Draw
257	Day 7140 to 7168	Health Check, Blood Draw
258	Day 7168 to 7196	Health Check, Blood Draw
259	Day 7196 to 7224	Health Check, Blood Draw
260	Day 7224 to 7252	Health Check, Blood Draw
261	Day 7252 to 7280	Health Check, Blood Draw
262	Day 7280 to 7308	Health Check, Blood Draw
263	Day 7308 to 7336	Health Check, Blood Draw
264	Day 7336 to 7364	Health Check, Blood Draw
265	Day 7364 to 7392	Health Check, Blood Draw
266	Day 7392 to 7420	Health Check, Blood Draw
267	Day 7420 to 7448	Health Check, Blood Draw
268	Day 7448 to 7476	Health Check, Blood Draw
269	Day 7476 to 7504	Health Check, Blood Draw
270	Day 7504 to 7532	Health Check, Blood Draw
271	Day 7532 to 7560	Health Check, Blood Draw
272	Day 7560 to 7588	Health Check, Blood Draw
273	Day 7588 to 7616	Health Check, Blood Draw
274	Day 7616 to 7644	Health Check, Blood Draw
275	Day 7644 to 7672	Health Check, Blood Draw
276	Day 7672 to 7700	Health Check, Blood Draw
277	Day 7700 to 7728	Health Check, Blood Draw
278	Day 7728 to 7756	Health Check, Blood Draw
279	Day 7756 to 7784	Health Check, Blood Draw
280	Day 7784 to 7812	Health Check, Blood Draw
281	Day 7812 to 7840	Health Check, Blood Draw
282	Day 7840 to 7868	Health Check, Blood Draw
283	Day 7868 to 7896	Health Check, Blood Draw
284	Day 7896 to 7924	Health Check, Blood Draw
285	Day 7924 to 7952	Health Check, Blood Draw
286	Day 7952 to 7980	Health Check, Blood Draw
287	Day 7980 to 8008	Health Check, Blood Draw
288	Day 8008 to 8036	Health Check, Blood Draw
289	Day 8036 to 8064	Health Check, Blood Draw
290	Day 8064 to 8092	Health Check, Blood Draw
291	Day 8092 to 8120	Health Check, Blood Draw
292	Day 8120 to 8148	Health Check, Blood Draw
293	Day 8148 to 8176	Health Check, Blood Draw
294	Day 8176 to 8204	Health Check, Blood Draw
295	Day 8204 to 8232	Health Check, Blood Draw
296	Day 8232 to 8260	Health Check, Blood Draw
297	Day 8260 to 8288	Health Check, Blood Draw
298	Day 8288 to 8316	Health Check, Blood Draw
299	Day 8316 to 8344	Health Check, Blood Draw
300	Day 8344 to 8372	Health Check, Blood Draw
301	Day 8372 to 8400	Health Check, Blood Draw
302	Day 8400 to 8428	Health Check, Blood Draw
303	Day 8428 to 8456	Health Check, Blood Draw
304	Day 84	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: - Geometric Mean Concentration (GMC) of Anti-Outer Surface Protein A (OspA) Immunoglobulin G (IgG) at 1 month after the fourth VLA15 dose. - Geometric Mean Fold Rise (GMFR) of Anti-OspA IgG concentrations from before the first dose to 1 month after the fourth VLA15 dose. **Measurement Method:** Central Laboratory Analysis of blood serum (~20 mL volume draws).

7.2 Secondary & Exploratory Endpoints - Percentage of participants reporting solicited and unsolicited adverse events. - Long-term seroconversion rates and duration of detectable antibody titers.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient monitoring via 8 in-clinic health checks and 3 remote phone follow-ups; tracking of local (injection site) and systemic reactions. **Serious Adverse Events (SAEs):** Standard expedited reporting timelines (typically 24 hours). **Data Monitoring Committee (DMC):** External Data Monitoring Committee (EDMC), comprised solely of external members independent of the Pfizer study team.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary immunogenicity outcomes; Safety Set for all participants who receive at least one dose. **Sample Size Justification:** Approximately $N=200$ (50 per arm) to ensure adequate power to compare immunogenicity and safety signals across the 4 differing schedules. **Handling of Missing Data:** Standard imputation methods for immunological data or as dictated by the pre-specified SAP.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Rigorous onsite and remote monitoring facilitated by the Sponsor; oversight by the independent EDMC. **Audit Trail:** Secure database for eCRF inputs, tracking of blinding protocols, and stringent third-party unblinding controls.

11. Study Identifiers & Governance

Primary ID: C4601019 **Registry Number:** NCT07226882 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study to Learn About Different Dosing Schedules of a Lyme Disease Vaccine in Healthy Adults **Official Regulatory Title:** A Phase 3, randomized, parallel-group, placebo-controlled, double-blinded (third-party unblinded) study to evaluate the safety, tolerability, and immunogenicity of different vaccination schedules of the 6-valent OspA-based Lyme disease (LD) vaccine, VLA15, in healthy participants 18 through 44 years of age.

12. Clinical Framework (Vaccine Specific)

Primary Condition: Lyme Disease (Prevention) **Diagnosis Threshold:** Healthy Participants (Pre-exposure Prophylaxis) **Medical Methodology:** Active Immunization (6-valent OspA-based vaccine) **Optimization Target:** Optimal vaccination dosing schedule maximizing immunogenicity and minimizing adverse events.

13. Study Procedures & Methodology

Management Pathway: Prophylactic vaccination pathway in a healthy population. **Education Protocol (HCP):** Blinded administration training and third-party unblinding procedures for clinical personnel. **Education Protocol (Patient):** Contraception compliance, AE recognition, and instructions for reporting reactions. **Quality Audit Mechanism:** Periodic EDMC charter reviews and compliance audits for adherence to randomized dosing schedules.

14. Observation & Follow-Up Schedule

Total Duration: ~24 months (2 years) **Onsite Visit Cadence:** 8 physical clinic visits encompassing health checks, blood draws, and injections. **Remote Monitoring Frequency:** 3 structured phone visits throughout the 2-year timeline. **Checklist Review Interval:** Continuous AE evaluation specifically post-vaccination intervals.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				